

DATA 266 Homework 2.5

DATA 266 — Homework 2.5

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SID4 = 0670 | SEED = 670 | SLICE = 670 | HP_ID = 4 | CLS_A = 0 | CLS_B = 5

HP_ID is only printed. I did not train a second net with a different lr or width.

I ran this on an RTX 4090 in the GPU lab (hostname ADS-R15-840-01). UUID GPU-5b052ad1-4272-40db-4b25-c930bf32b547. Scripts: run_quick.bat then run_full.bat. Every number below is from that UUID.

Part A — the card

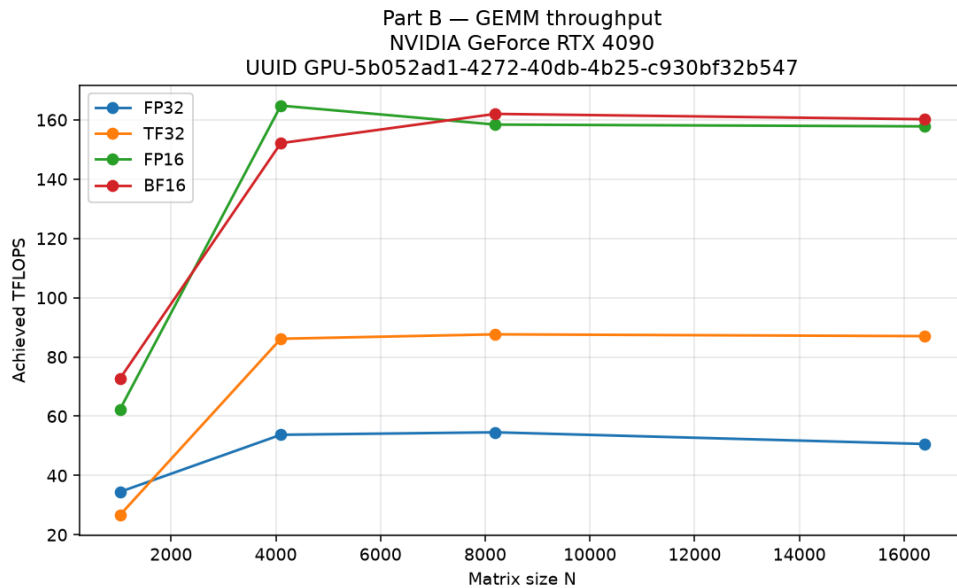
My GPU	
Name	NVIDIA GeForce RTX 4090
UUID	GPU-5b052ad1-4272-40db-4b25-c930bf32b547
Driver	595.95
CUDA	13.2
VRAM	24564 MiB
Power limit	450.00 W
Architecture	Ada Lovelace (AD102)
Memory	24 GB GDDR6X, 1008 GB/s
Tensor cores	4th gen; TF32, FP16, BF16, FP8, INT8, INT4

I copied the nvidia-smi lines from results/NVIDIA GeForce RTX 4090_5b052ad1/nvidia_smi_q.txt. Architecture / bandwidth / tensor cores are from the NVIDIA Ada GPU Architecture

whitepaper: <https://images.nvidia.com/aem-dam/Solutions/geforce/ada/nvidia-ada-gpu-architecture.pdf>. Hours: [reservations/GPU_HOURS.md](#).

Part B — precision and throughput

I multiplied square matrices at $N = 1024, 4096, 8192, 16384$ in FP32, TF32, FP16, and BF16. Warmup first, then enough repeats that the times settled. Best BF16 I got was **162 TFLOPS** at $N=8192$, which is **98%** of the 165.2 peak in the whitepaper.



GEMM TFLOPS vs matrix size

Small N never looks like the spec. At 1024 the kernel is tiny, setup time is a big chunk of the clock, and the matrices still fit in cache. FP16 levels off around $N=4096$ (164.8 TFLOPS). BF16 needs $N=8192$ to get to 162. FP32 tops out around 55 TFLOPS.

N	precision	TFLOPS	% of peak	reps	mean ms
1024	fp32	34.47	41.7	200	0.062
4096	fp32	53.74	65.1	200	2.558
8192	fp32	54.57	66.1	75	20.150
16384	fp32	50.63	61.3	20	173.731
1024	tf32	26.67	32.3	200	0.081
4096	tf32	86.13	104.3	200	1.596
8192	tf32	87.62	106.1	120	12.548

N	precision	TFLOPS	% of peak	reps	mean ms
16384	tf32	87.02	105.3	20	101.086
1024	fp16	62.31	37.7	200	0.034
4096	fp16	164.80	99.8	200	0.834
8192	fp16	158.40	95.9	200	6.941
16384	fp16	157.80	95.5	27	55.741
1024	bf16	72.71	44.0	200	0.030
4096	bf16	152.14	92.1	200	0.903
8192	bf16	162.00	98.1	200	6.787
16384	bf16	160.21	97.0	28	54.905

TF32 went a bit over 100% of 82.6. The whitepaper uses a 2520 MHz boost. This card's max graphics clock in nvidia-smi is 3105 MHz, so I am not treating that as a bug.

I also tried a lower precision. FP8 ran: **263.65 TFLOPS** at N=8192. FP4 did not. PyTorch had a dtype name (float4_e2m1fn_x2) and no multiply I could call. This 4090 does not have FP4 tensor cores anyway.

Part C — bandwidth vs compute

I ran a big elementwise add and a big BF16 matmul.

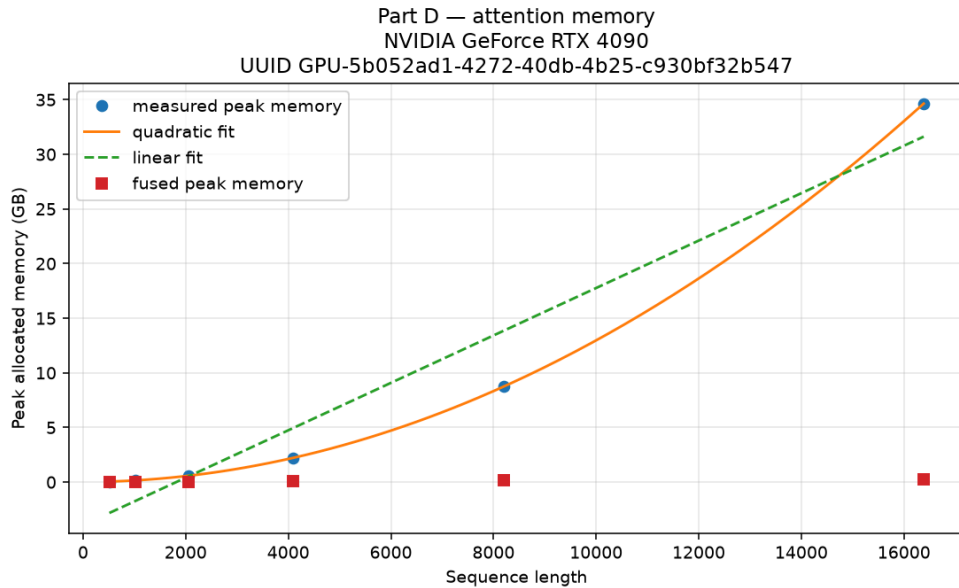
Add: **915.1 GB/s**, **90.8%** of 1008. Arithmetic intensity is **0.083 FLOP/byte** (1 add, 12 bytes). That is left of the ridge (~164 FLOP/byte for BF16 on this card), so memory-bound. A copy was 908.5 GB/s.

Large BF16 GEMM (N=16384): 160.8 TFLOPS. Arithmetic intensity is **5461 FLOP/byte**. That is right of the ridge, so compute-bound.

The spec sheet bandwidth and the spec sheet TFLOPS are not what I actually got, and they are not the same kind of limit. Add never hit 1008 GB/s. GEMM never hit the sparse marketing number either.

Part D — cost of attention

I used batch 1, 32 heads, head size 64, bfloat16. Naive attention builds the full S×S matrix. Fused is scaled_dot_product_attention.



Peak memory vs sequence length

seq	naive ms	naive GB	fused ms	fused GB	speedup
512	0.10	0.050	0.13	0.025	0.82×
1024	0.45	0.160	0.31	0.034	1.48×
2048	1.89	0.579	1.08	0.050	1.75×
4096	7.23	2.223	2.61	0.084	2.77×
8192	28.67	8.733	4.40	0.151	6.51×
16384	918.11	34.637	17.46	0.285	52.59×

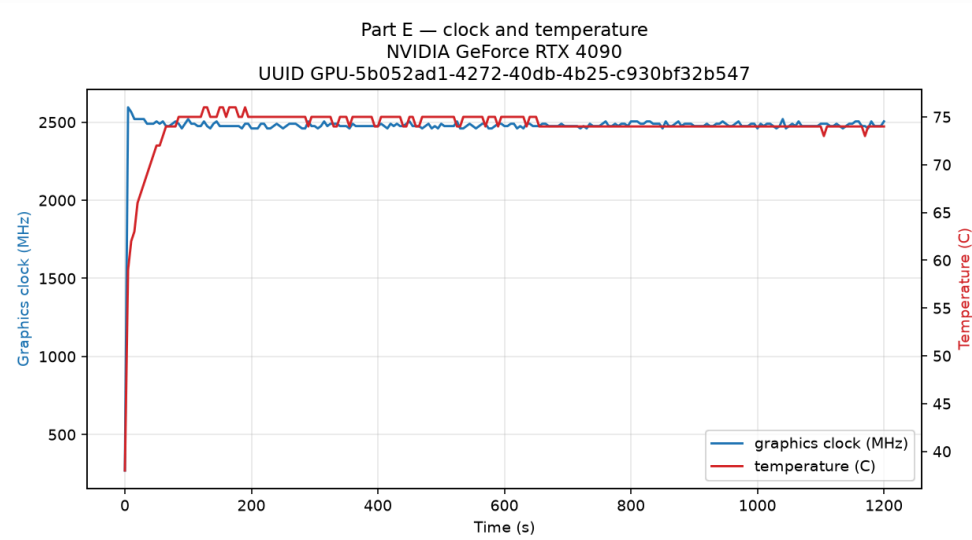
Naive still ran at 16384, but peak memory was 34.6 GB on a 24 GB card (Windows shared memory). It died later: last length that worked **26752**, first that failed **26816**. I did not search one token at a time. Fused did not run out of memory up to 131072.

I fit peak memory vs S. Quadratic beat linear. The S^2 coefficient was **128 bytes per token²**. That is the $S \times S$ matrix.

Fused never writes the $S \times S$ score matrix into GPU memory. It loads Q/K/V in tiles, does the products in SRAM, keeps a running softmax, and only writes the output tile. That cuts the $O(S^2)$ allocation and the $O(S^2)$ memory traffic. That is why memory stays small and why 16384 is 53× faster.

Part E — 20 minutes

I left a BF16 GEMM running for 20 minutes and sampled clock, temp, and power every 5 seconds.



Clock and temperature

Clock stayed around 2500 MHz. Temp went up to about **76 °C** and stayed there. Around 10 seconds in, power was already **450 W** (the limit). The clock did not drop. So this is hitting the power cap, not melting.

First 30 seconds: 155 TFLOPS. Last 5 minutes: 152 TFLOPS. **Steady / peak = 98%**. A 4 second run would have looked a bit faster. Training later will look like the 20 minute number.

Part F — Analysis

Table HW2.5.1. Every cell is from UUID GPU-5b052ad1-4272-40db-4b25-c930bf32b547 in RUN_LOG.txt (full run, not the smoke test).

Measurement	RTX 4090	Notes
Peak achieved TFLOPS (BF16)	162.00 (N=8192)	Part B
% of theoretical peak (BF16)	98.1% of 165.2	
Effective bandwidth (GB/s)	915.1 (90.8% of 1008)	add
Naive attention OOM length	worked at 26752, failed at 26816	

Measurement	RTX 4090	Notes
Fused attention OOM length	still worked at 131072	no fail in the search
Steady-state / peak throughput	98.0%	last 5 min / first 30 s
Throttle onset (s, or none)	~10 s, power 450 W	clock did not fall

Conclusion

On this 4090, big BF16 multiplies are close to the tensor peak (162 TFLOPS). A plain add only gets 91% of the memory spec. Those two numbers are what bound everything after week 3, and they do not match.

Naive attention grows like S^2 . At 16384 it was already 35 GB and 918 ms. Fused stayed under 0.3 GB and 18 ms. If I train a transformer later I need the fused kernel, and I should believe the 20 minute throughput, not a short benchmark.